Forecasting and Macroeconomic Analytics

ECON 522

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Today's Agenda

1. Simple and Multiple Linear Regression
2. OLS Estimation and Interpretation
3. Gauss-Markov Assumptions
4. Goodness of Fit: R^2 and Adjusted R^2
5. Hypothesis Testing (t-tests and F-tests)
6. Sampling Distribution and Inference
7. Small Sample vs. Asymptotic Properties
8. Dummy Variables and Interactions

The Linear Regression Model

Running Example: Housing Prices

Simple Linear Regression Model

Throughout this lecture, we'll use a running example:

Goal: Predict house prices based on characteristics

Variables:

- Y: House price (thousands of dollars)
- X_1: Square footage (100s of sq ft)
- X_2: Number of bedrooms
- X_3: Age of house (years)
- D: Dummy for waterfront property (1=yes, 0=no)

We'll start simple and build complexity as we go.

Simple Linear Regression Model

The population model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

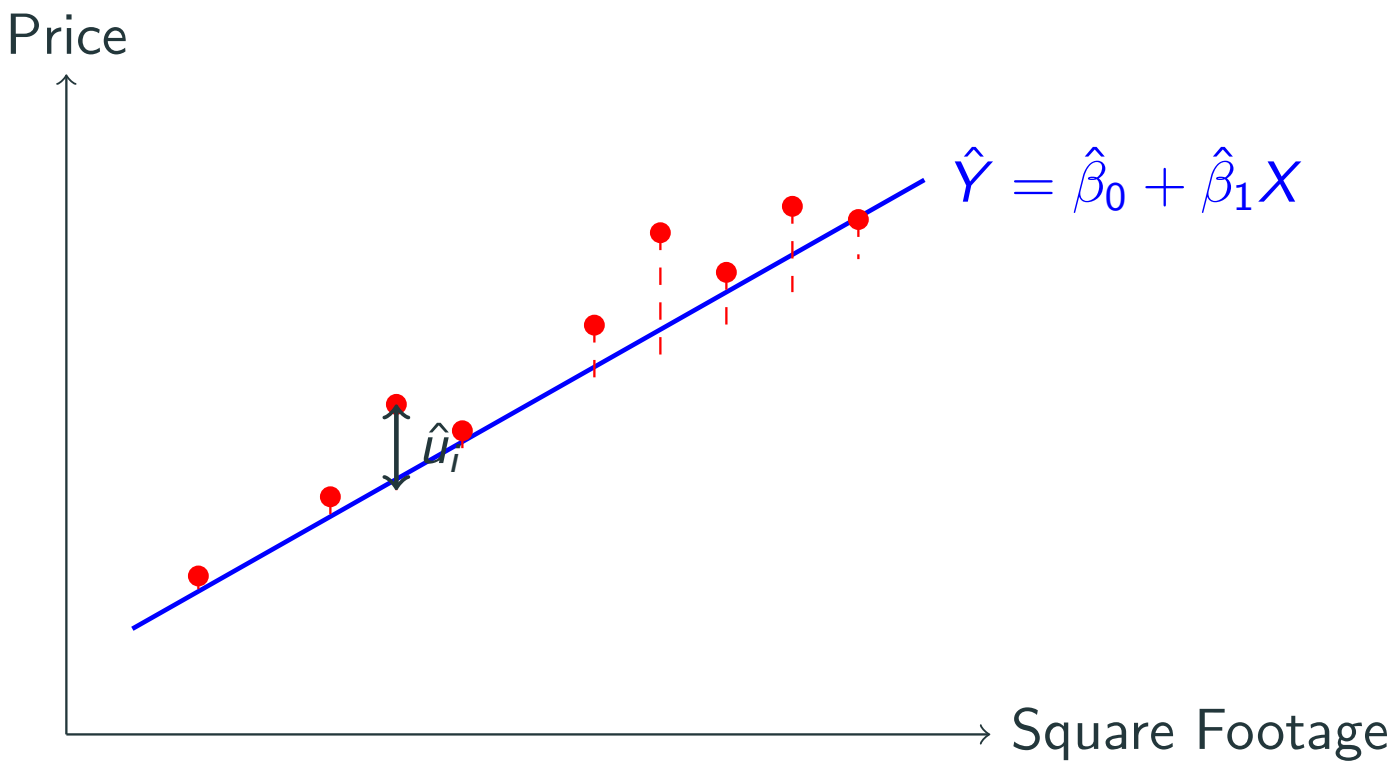
where:

- Y_i: dependent variable (house price)
- X_i: independent variable (square footage)
- β_0: intercept (price when $X = 0$)
- β_1: slope (change in Y for one unit change in X)
- u_i: error term (unobserved factors)

population parameters

Key insight: We never observe β_0, β_1 or u_i. We must estimate them from data.

Visual Representation



The red dashed lines show residuals: $\hat{u}_i = Y_i - \hat{Y}_i$

Multiple Linear Regression

Extending to multiple predictors:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

For our housing example:

$$\text{Price}_i = \beta_0 + \beta_1 \text{SqFt}_i + \beta_2 \text{Bedrooms}_i + \beta_3 \text{Age}_i + u_i$$

Interpretation of β_j:

- The change in Y for a one-unit increase in X_j, *holding all other variables constant* (ceteris paribus)
- β_1: Price increase for 100 additional sq ft, holding bedrooms and age constant

OLS Estimation

The OLS Principle

Ordinary Least Squares chooses $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k$ to minimize the sum of squared residuals: your "miss" for each observation

$$\min_{\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k} \sum_{i=1}^{n} \hat{u}_i^2 = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$$

where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \cdots + \hat{\beta}_k X_{ki}$

#square function convenient for some but not others. For example, with stocks you might be happier with overperformance rather than underperformance

Why squared residuals?

- Treats positive and negative errors symmetrically
- Penalizes large errors more than small ones
- Mathematically tractable (differentiable)

(convenience:

OLS Formulas: Simple Regression

For simple regression $Y_i = \beta_0 + \beta_1 X_i + u_i$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

Intuition:

- $\hat{\beta}_1$ measures how X and Y co-move, scaled by variation in X
- The regression line always passes through $(\bar{X}, \bar{Y})$

- You must have enough variation to lean relationship between X and Y

Example: Python Output (Simple Regression)

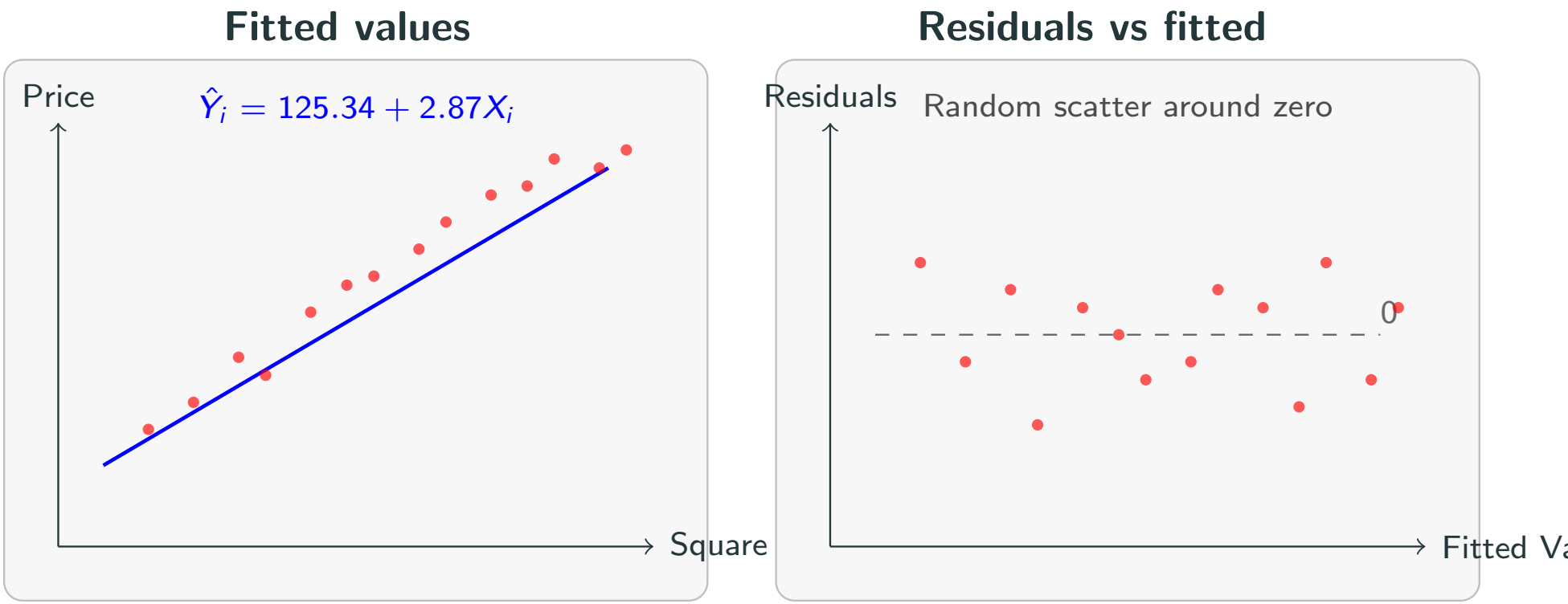
square footage explains 62% of variation

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.623
Model:                            OLS   Adj. R-squared:                  0.621
No. Observations:                 200   F-statistic:                     327.4
Df Residuals:                     198   Prob (F-statistic):           2.15e-44
Df Model:                           1
==============================================================================
                 coef    std err          t      P>|t|      [0.025  0.975]
------------------------------------------------------------------------------
const        125.3421      8.234     15.223      0.000     109.093  141.591
sqft           2.8654      0.158     18.094      0.000       2.553    3.178
==============================================================================
```

Interpretation: Each additional 100 sq ft increases price by $2,865.40, on average.

Fitted Values and Residuals Plot

Need absense of serial correlation



Left: fitted line and data. Right: residuals should show no pattern.

* a good fit would have a random pattern of plotted residuals vs fitted

Gauss-Markov Assumptions

The Classical Assumptions

time series is not independent — violates A3

For OLS to have desirable properties, we assume:

1. **Linearity:** $Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + u_i$
2. **Random sampling:** $(Y_i, X_{1i}, \ldots, X_{ki})$ are i.i.d. IID
3. **No perfect collinearity:** No X_j is a perfect linear combination of others
4. **Zero conditional mean:** $E[u_i | X_{1i}, \ldots, X_{ki}] = 0$ (exogeneity)
5. **Homoskedasticity:** $\text{Var}(u_i | X_{1i}, \ldots, X_{ki}) = \sigma^2$ → spread of errors is constant
6. **Normality:** $u_i | X_{1i}, \ldots, X_{ki} \sim N(0, \sigma^2)$ (for exact inference)

Under assumptions 1-4: OLS is **unbiased** and **consistent**

Under assumptions 1-5: OLS is **BLUE** (Best Linear Unbiased Estimator)

Key Assumption: Exogeneity

Zero Conditional Mean: $E[u_i|X_i] = 0$

This means X_i and u_i are uncorrelated and X_i does not contain information about u_i.

omitted variable bias doesn't matter for forecasting because we don't care about the "why"

When does this fail?

- **Omitted variable bias:** Leaving out important variables
- **Measurement error:** Error in measuring X – matters
- **Simultaneity:** Y also affects X
- **Sample selection:** Non-random sample

Example in housing: If we omit neighborhood quality (affects both price and size choices), $E[u_i|\text{SqFt}_i] \neq 0$ and $\hat{\beta}_1$ is biased.

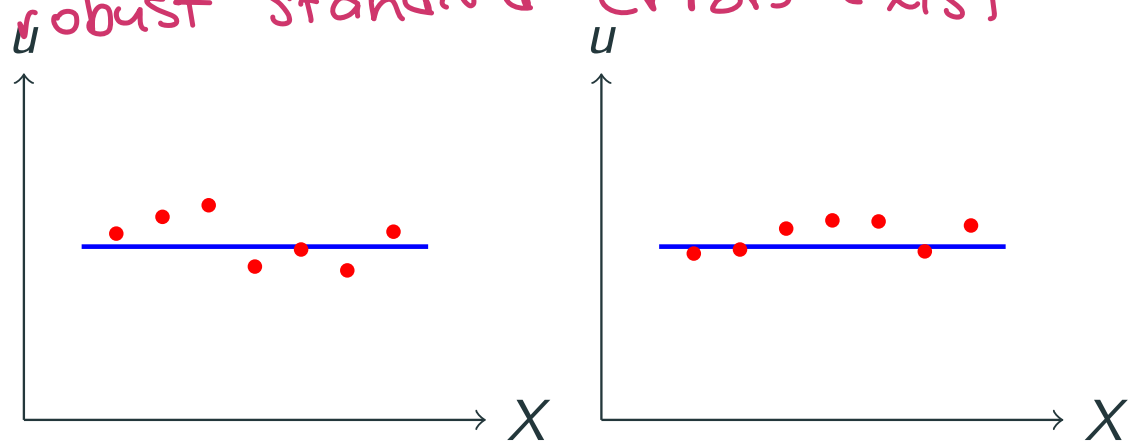
Heteroskedasticity

Homoskedasticity: $\text{Var}(u_i|X_i) = \sigma^2$ (constant variance) what you want

Heteroskedasticity: $\text{Var}(u_i|X_i) = \sigma_i^2$ (variance depends on X)

↳ affects standard error, not biasedness.

↳ why robust standard errors exist



Homoskedastic Heteroskedastic

Consequence: OLS still unbiased but standard errors are wrong! Use robust (Huber-White) standard errors.

Serial Correlation (Time Series)

No serial correlation: $\text{Cov}(u_i, u_j) = 0$ for $i \neq j$

Serial correlation: Errors in different observations are correlated

Common in time series:

- Omitted trends
- Omitted dynamics (lagged effects)
- Persistent shocks

: If last month's interest rate affects this month's prices, but you only include current interest rates, that lagged effect lives in the error term and creates correlation.

Persistent shocks: A recession hits in month 5. It doesn't fully resolve by month 6. The "recession effect" in u_t spills into u_{t+1}.

Consequence:

- OLS still unbiased
- Standard errors are wrong (usually too small)
- Use Newey-West HAC standard errors
- Important for AR models we'll study later!

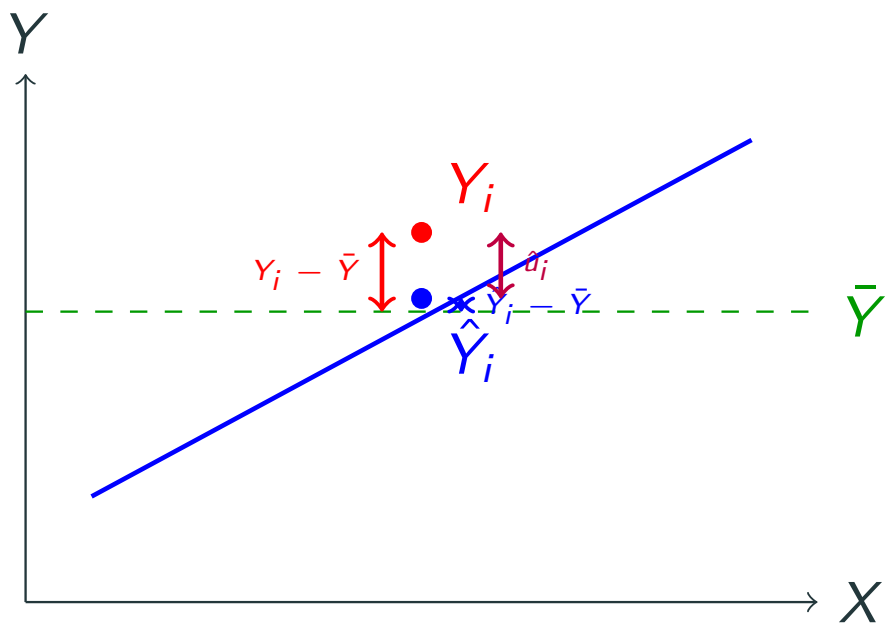
The usual standard error formula assumes each observation provides independent information. But if errors are positively correlated, your 200 observations aren't really 200 independent pieces of information—there's redundancy. You're overstating your precision, so confidence intervals are too narrow and you reject nulls too often.

Goodness of Fit

Decomposing Variation

Total variation in Y can be decomposed:

$$\underbrace{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}_{\text{Total Sum of Squares (TSS)}} = \underbrace{\sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2}_{\text{Explained Sum of Squares (ESS)}} + \underbrace{\sum_{i=1}^{n}\hat{u}_i^2}_{\text{Residual Sum of Squares (RSS)}}$$



R^2: Coefficient of Determination

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{\sum \hat{u}_i^2}{\sum (Y_i - \bar{Y})^2}$$

R^2 = ESS/TSS is just asking: what fraction of the total variation did we explain?

Interpretation:

- Fraction of variation in Y explained by the model
- Ranges from 0 to 1
- $R^2 = 0.623$ means 62.3% of price variation is explained by square footage

example: regress number of churches on number of bars across cities. High R^2! Both are driven by population, which you omitted. Spurious correlation can produce great "fit."

Important caveats:

- R^2 always increases when adding variables (even irrelevant ones!)
- High R^2 doesn't mean good model (could be spurious)
- Low R^2 doesn't mean bad model (individual predictions may be poor, but coefficients still meaningful) – You just need coefficient to be unbiased

Adjusted R^2

Adjusted R^2 penalizes adding variables:

$$\bar{R}^2 = 1 - \frac{\text{RSS}/(n-k-1)}{\text{TSS}/(n-1)} = 1 - (1 - R^2)\frac{n-1}{n-k-1}$$

where k is the number of regressors (excluding constant).

Key differences from R^2:

- Adjusts for degrees of freedom
- Can decrease when adding variables
- Better for comparing models with different numbers of predictors
- Always $\bar{R}^2 \leq R^2$

Rule of thumb: Use $\bar{R}^2$ for model comparison, R^2 for interpretation.

Hypothesis Testing

Sampling Distribution of OLS Estimators

$\hat{\beta}_j$ is a random variable (depends on random sample).

Under Gauss-Markov assumptions:

- $E[\hat{\beta}_j] = \beta_j$ (unbiased)
- $\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}$ where:
  - $SST_j = \sum(X_{ji} - \bar{X}_j)^2$ (variation in X_j)
  - R_j^2 is R^2 from regressing X_j on all other X's

Standard error: $SE(\hat{\beta}_j) = \sqrt{\widehat{\text{Var}}(\hat{\beta}_j)}$

With normality assumption:

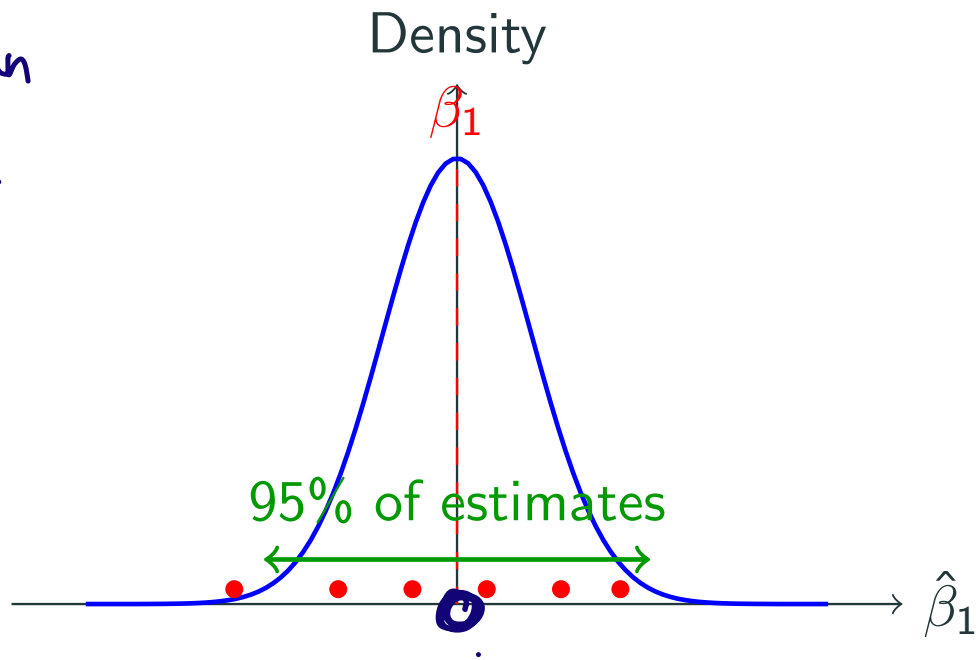
$$\hat{\beta}_j \sim N\left(\beta_j, \text{Var}(\hat{\beta}_j)\right)$$

Visualizing the Sampling Distribution

What does it mean to be unbiased?

$E[\hat{\beta}] = \beta$

↑ mean



Different samples give different $\hat{\beta}_1$

$\longleftrightarrow$
$SE(\hat{\beta}_1)$

In repeated samples, $\hat{\beta}_1$ centers around true β_1 with spread $SE(\hat{\beta}_1)$.

t-Statistic and Hypothesis Testing

To test $H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$:

$$t = \frac{\hat{\beta}_j - 0}{SE(\hat{\beta}_j)} \sim t_{n-k-1}$$

Decision rule (5% significance level):

- If $|t| > 1.96$ (or critical value): Reject H_0, conclude $\beta_j \neq 0$
- If $|t| < 1.96$: Fail to reject H_0
- Or use p-value: Reject if $p < 0.05$

From our example:

- $\hat{\beta}_1 = 2.8654$, $SE = 0.158$
- $t = 2.8654/0.158 = 18.09$
- $p < 0.001$ (highly significant)

Confidence Intervals

A 95% confidence interval for β_j:

$$\hat{\beta}_j \pm 1.96 \times SE(\hat{\beta}_j)$$

(Use t_{n-k-1} critical value for small samples instead of 1.96)

Interpretation:

- If we repeated sampling many times, 95% of intervals would contain true β_j
- NOT: "95% probability that β_j is in this interval" (frequentist view)

From our example:

$$[2.553, 3.178] = 2.8654 \pm 1.96 \times 0.158$$

We're 95% confident the true effect is between \$2,553 and \$3,178 per 100 sq ft.

F-Test: Overall Significance

Test if *any* of the variables matter: $H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$

$$F = \frac{(TSS - RSS)/k}{RSS/(n-k-1)} = \frac{R^2/k}{(1-R^2)/(n-k-1)} \sim F_{k,n-k-1}$$

Decision rule:

- If $F > F_{critical}$: Reject H_0, model is significant
- Common cutoff: $F > 3$ or 4 (rule of thumb)

From our example: $F = 327.4$ with $p < 0.001$

Strongly reject that all coefficients are zero.

Note: Can also use F-test for testing multiple restrictions jointly.

Small Sample vs. Asymptotics

Exact (Small Sample) Properties

Finite sample properties hold for any sample size n:

Under Gauss-Markov assumptions (1-5):

- $E[\hat{\beta}_j] = \beta_j$ (unbiased for any n)
- OLS is BLUE (minimum variance among linear unbiased estimators)
- With normality (assumption 6): t-statistics follow t_{n-k-1} exactly

Implications:

- Hypothesis tests are valid even with $n = 30$
- But: estimates may be imprecise (large standard errors)
- Power of tests increases with n

Asymptotic (Large Sample) Properties

Asymptotic properties hold as $n \to \infty$:

You can relax assumptions as n gets large

Consistency: $\hat{\beta}_j \xrightarrow{p} \beta_j$ as $n \to \infty$

- Estimates converge to true values
- Requires only assumptions 1-4 (weaker than unbiasedness!)

Asymptotic normality:

$$\sqrt{n}(\hat{\beta}_j - \beta_j) \xrightarrow{d} N(0, V)$$

- Don't need normality assumption
- t-statistics approximately $N(0, 1)$ for large n
- Justifies using $z = 1.96$ instead of t_{n-k-1}

Practical implication: With $n > 100$, can relax normality assumption.

Small vs. Large Sample: Summary

Property	Small Sample	Large Sample
Unbiasedness	Exact (with assumptions 1-4)	Exact
BLUE	Exact (with assumptions 1-5)	Approximately
Distribution	Need normality for t-dist	Approx. normal (CLT)
Inference	t_{n-k-1} critical values	z critical values (1.96)
Heteroskedasticity	Biased SEs, use robust	Robust SEs valid

Practical guidelines:

- $n < 30$: Need normality, use t-distribution carefully
- $30 < n < 100$: Use t-distribution, robust SEs if suspect heteroskedasticity
- $n > 100$: Asymptotic approximations work well, always use robust SEs

Dummy Variables

Introduction to Dummy Variables

Dummy variable: Binary indicator (0 or 1)

many things change with seasons

Examples:

- Gender: $D = 1$ if female, $D = 0$ if male
- Treatment: $D = 1$ if treated, $D = 0$ if control
- Region: $D = 1$ if urban, $D = 0$ if rural
- In our example: $D = 1$ if waterfront property, $D = 0$ otherwise

Why use dummies?

- Capture qualitative differences between groups
- Allow intercept to differ across groups
- Test for structural differences

* Any retail sale from FRED

Single Dummy Variable Model

$$\text{Price}_i = \beta_0 + \beta_1 \text{SqFt}_i + \beta_2 \text{Waterfront}_i + u_i$$

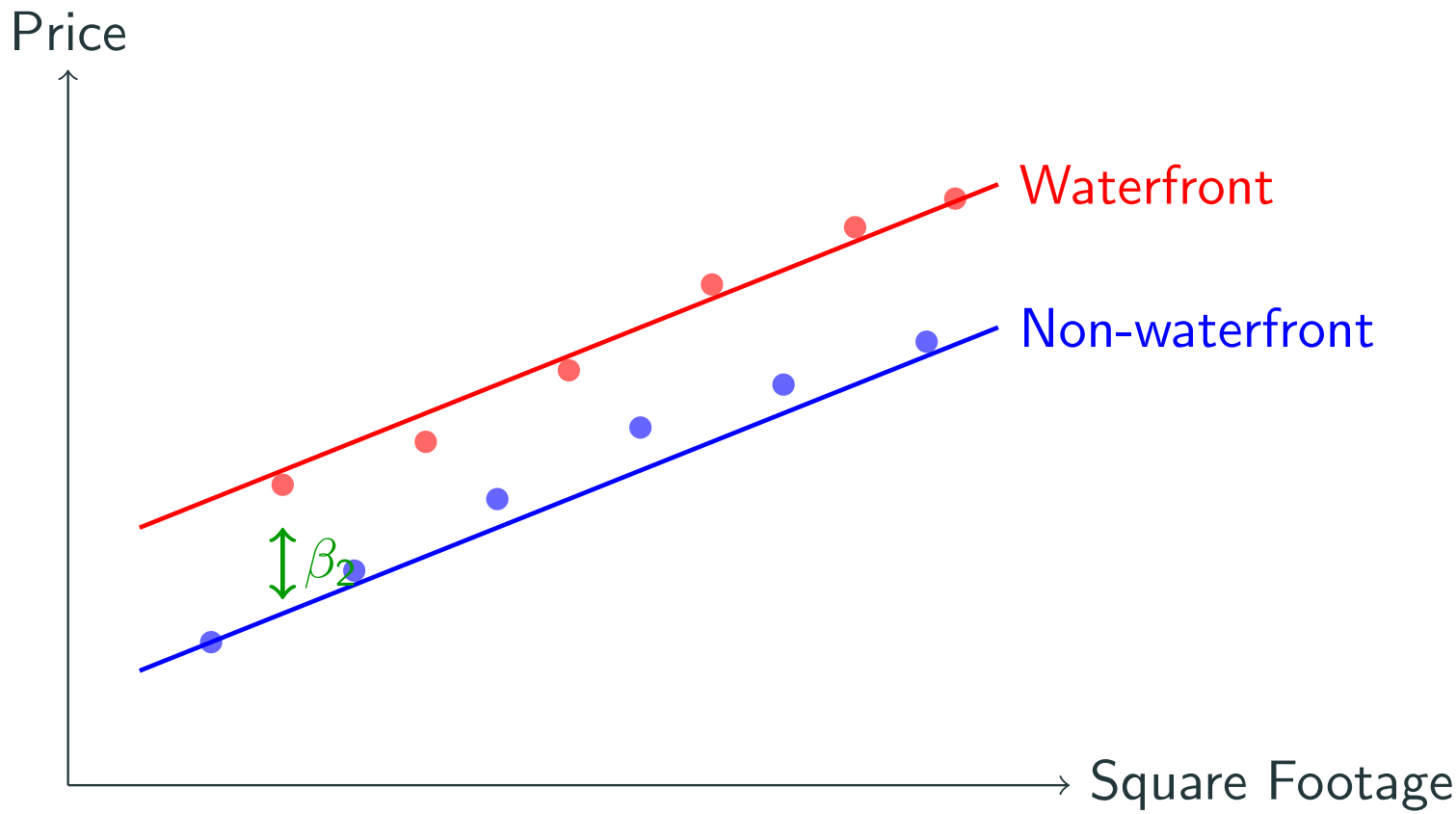
Interpretation:

- β_0: Intercept for non-waterfront properties ($D = 0$)
- $\beta_0 + \beta_2$: Intercept for waterfront properties ($D = 1$)
- β_2: Waterfront premium (difference in price, holding sqft constant)
- β_1: Effect of sqft (same slope for both groups)

Two parallel regression lines:

- Non-waterfront: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$
- Waterfront: $\hat{Y} = (\hat{\beta}_0 + \hat{\beta}_2) + \hat{\beta}_1 X$

Visual: Single Dummy Variable



Same slope β_1, different intercepts

Example: Python Output with Dummy

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  price   R-squared:                       0.741
Model:                            OLS   Adj. R-squared:                  0.738
No. Observations:                 200
==============================================================================
                 coef    std err          t      P>|t|      [0.025  0.975]
------------------------------------------------------------------------------
const        118.2341      7.123     16.602      0.000     104.194  132.274
sqft           2.6543      0.142     18.691      0.000       2.374    2.934
waterfront    87.3421      9.856      8.862      0.000      67.903  106.781
==============================================================================
```

holding waterfront constant

Interpretation:

- Waterfront properties are \$87,342 more expensive on average, holding sqft constant
- Highly significant ($t = 8.86$, $p < 0.001$)

Multiple Categories: More Than Two Groups

For a categorical variable with m categories, use $m - 1$ dummies (avoid dummy variable trap!)

Example: House type (3 categories)

- Single-family (base category)
- Condo: $D_1 = 1$ if condo, 0 otherwise
- Townhouse: $D_2 = 1$ if townhouse, 0 otherwise

$$\text{Price}_i = \beta_0 + \beta_1 \text{SqFt}_i + \beta_2 D_{1i} + \beta_3 D_{2i} + u_i$$

Interpretation:

You can't have constant + dummy variable, so you have to leave out 1, pick 1 as base category

- β_0: Intercept for single-family homes (base)
- β_2: Condo premium/discount vs. single-family
- β_3: Townhouse premium/discount vs. single-family
- Condo vs. townhouse difference: $\beta_2 - \beta_3$ (requires F-test)

The Dummy Variable Trap

NEVER include all m dummies plus a constant!

$$\text{Price}_i = \beta_0 + \beta_1 D_{\text{single}} + \beta_2 D_{\text{condo}} + \beta_3 D_{\text{town}} + u_i$$

↳ inc

Problem: Perfect multicollinearity!

- $D_{\text{single}} + D_{\text{condo}} + D_{\text{town}} = 1$ (always)
- Cannot separately identify $\beta_0, \beta_1, \beta_2, \beta_3$
- Computer will drop one variable automatically or give error

Solution: Either

1. Include constant + $(m-1)$ dummies (most common)
2. Include m dummies, no constant (less intuitive)

Interaction Terms: Different Slopes

What if the effect of sqft differs between waterfront and non-waterfront?

$$\text{Price}_i = \beta_0 + \beta_1 \text{SqFt}_i + \beta_2 \text{Waterfront}_i + \beta_3(\text{SqFt}_i \times \text{Waterfront}_i) + u_i$$

↳ incremental effect of square feet for being on waterfront

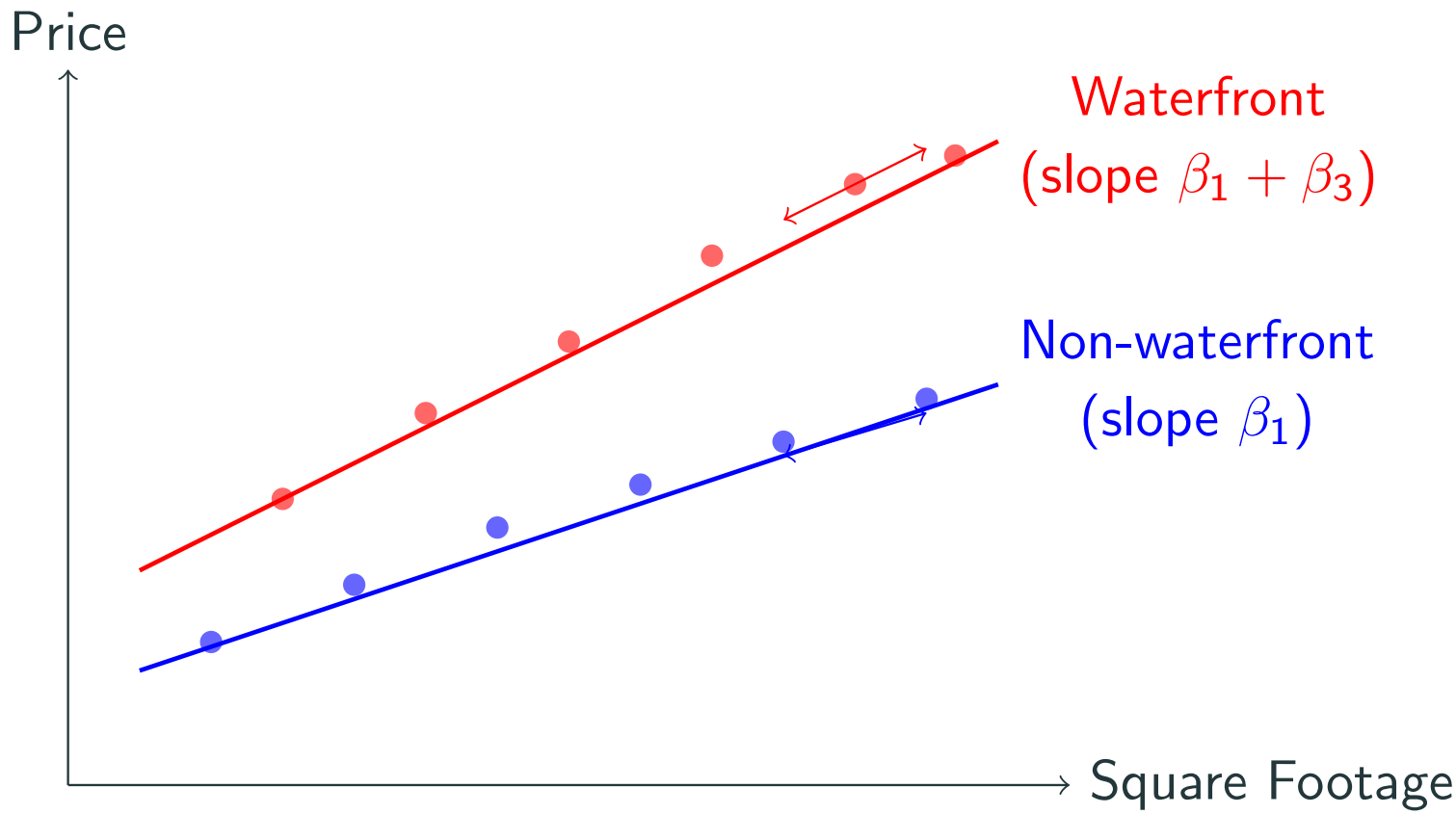
Two separate regression lines:

- Non-waterfront ($D = 0$): $\text{Price} = \beta_0 + \beta_1 \text{SqFt}$
- Waterfront ($D = 1$): $\text{Price} = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)\text{SqFt}$

Interpretation of β_3:

- Difference in slopes between groups
- Additional value of sqft for waterfront properties
- If $\beta_3 > 0$: sqft matters more for waterfront homes

Visual: Interaction Effect



Different slopes AND intercepts

Now both the level AND the effect of sqft differ by waterfront status.

Testing the Interaction

Question: Do we need the interaction term?

Test $H_0 : \beta_3 = 0$ (slopes are the same)

- Use t-test on $\hat{\beta}_3$
- If reject: slopes differ significantly, keep interaction
- If fail to reject: simpler model (parallel lines) is adequate

Alternative: F-test comparing models

- Restricted model: No interaction (parallel lines)
- Unrestricted model: With interaction
- $F = \frac{(RSS_R - RSS_{UR})/q}{RSS_{UR}/(n-k-1)}$ where q is number of restrictions

Continuous $\times$ Continuous Interactions

Interactions aren't just for dummies!

$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 (X_{1i} \times X_{2i}) + u_i$

Example: $\text{Price}_i = \beta_0 + \beta_1 \text{SqFt}_i + \beta_2 \text{Age}_i + \beta_3 (\text{SqFt}_i \times \text{Age}_i) + u_i$

Marginal effects:

- $\frac{\partial \text{Price}}{\partial \text{SqFt}} = \beta_1 + \beta_3 \text{Age}$
- Effect of sqft depends on age of house
- If $\beta_3 < 0$: sqft matters less for older houses (depreciation)

Interpretation is more complex: Main effects are conditional on other variable $= 0$.

Common Mistakes with Dummies

1. **Dummy variable trap**: Including all categories + constant
   - Solution: Drop one category (make it the base)
2. **Wrong interpretation**: Thinking β_2 is effect for $D = 1$ group
   - Correct: β_2 is difference between groups
3. **Forgetting interactions**: Assuming parallel slopes when they differ
   - Test for interactions if theory suggests different effects
4. **Arbitrary base category**: Comparing to wrong reference
   - Choose meaningful base (e.g., control group, most common category)
5. **Too many dummies**: Creating dummies for rare categories
   - Group small categories together

Practical Considerations

Model Specification Issues

How do we choose which variables to include?

Theory-driven approach:

- Start with economic/theoretical model
- Include variables that theory suggests are important
- Avoid data mining (testing hundreds of specifications)

Common specification errors:

1. **Omitted variable bias**: Leaving out important variables
   - Bias: $E[\hat{\beta}_1] = \beta_1 + \beta_2 \frac{\text{Cov}(X_1, X_2)}{\text{Var}(X_1)}$
   - Only unbiased if omitted variable uncorrelated with included ones
2. **Overcontrolling**: Including variables affected by treatment
3. **Functional form**: Wrong relationship (linear vs. log, quadratic, etc.)

Functional Forms

Not all relationships are linear!

Log stabilizes variance

Common transformations:

- **Log-log:** $\log(Y) = \beta_0 + \beta_1 \log(X) + u$
  - β_1 is elasticity (percentage change interpretation)
- **Log-level:** $\log(Y) = \beta_0 + \beta_1 X + u$
  - $\beta_1 \times 100$ is semi-elasticity (percentage change in Y per unit X)
- **Level-log:** $Y = \beta_0 + \beta_1 \log(X) + u$
  - $\beta_1/100$ is change in Y per 1% change in X
- **Quadratic:** $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + u$
  - Allows for non-monotonic relationships

Check: Plot residuals vs. fitted values to detect misspecification.

Diagnostics: What to Check

After estimating your model, always check:

1. **Residual plots**
   - Plot $\hat{u}_i$ vs. $\hat{Y}_i$ or vs. each X_j
   - Look for patterns (suggests misspecification)
   - Look for heteroskedasticity (funnel shape)
2. **Normality of residuals**
   - Histogram or Q-Q plot
   - Less important with large n (CLT applies)
3. **Outliers and influential observations**
   - Check leverage, Cook's distance
   - One outlier can dramatically change results
4. **Multicollinearity**
   - Check VIF (Variance Inflation Factor)
   - High VIF (¿10) means highly correlated predictors

Connection to Time Series

Everything we covered applies to time series, but with important caveats:

Why OLS matters for AR models:

- AR(1): $Y_t = \beta_0 + \beta_1 Y_{t-1} + u_t$ looks like OLS regression!
- Same estimation principle (minimize squared residuals)
- Similar interpretation of coefficients

Key differences in time series:

- **Serial correlation**: Errors often correlated over time (violates i.i.d.)
- **Non-stationarity**: Mean/variance may change over time
- **Dynamics**: Lagged dependent variables common (create new issues)
- **Standard errors**: Need to account for autocorrelation (Newey-West)

Understanding OLS is the foundation for time series econometrics!

Summary

→ remember!

→ student life

Key Takeaways

1. **OLS minimizes squared residuals**, yielding best linear fit
2. **Under Gauss-Markov assumptions**, OLS is unbiased and BLUE
3. **Violations matter**:
   - Heteroskedasticity/serial correlation $\rightarrow$ wrong SEs (use robust)
   - Omitted variables $\rightarrow$ biased estimates
4. R^2 measures fit; use **adjusted** R^2 for model comparison
5. **t-tests** for individual coefficients, **F-tests** for joint hypotheses
6. **Asymptotic theory** relaxes normality, needs large n
7. **Dummies** capture group differences; use $m - 1$ dummies for m groups
8. **Interactions** allow effects to vary across groups or levels

Checklist for Running Regressions

Before running:

- Clear research question and theoretical model
- Check for missing data
- Plot relationships (scatterplots)

After running:

- Check significance (t-stats, p-values)
- Examine R^2 and adjusted R^2
- Plot residuals (look for patterns)
- Check for outliers
- Use robust standard errors if needed
- Interpret coefficients in context (with units!)

Reporting:

- Always report SEs or t-stats (not just coefficients)
- Report n, R^2, and F-statistic

Resources and Next Steps

For deeper understanding:

- Wooldridge (2020): *Introductory Econometrics*, Chapters 2-7
- Greene (2018): *Econometric Analysis* (more advanced)
- Stock & Watson (2020): *Introduction to Econometrics*

Python resources:

- `statsmodels`: Full-featured econometric modeling
- `scikit-learn`: Machine learning perspective on regression
- `linearmodels`: Advanced panel and IV methods

Coming up in our course:

- AR, MA, ARMA models (apply OLS principles to time series)
- Stationarity and unit roots
- Forecasting
- Vector autoregressions (VAR)

Questions?

Appendix: Derivation of OLS Estimator

Minimize: $S(\beta_0, \beta_1) = \sum_{i=1}^{n}(Y_i - \beta_0 - \beta_1 X_i)^2$

First-order conditions: $\frac{\partial S}{\partial \beta_0} = -2\sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$

$\frac{\partial S}{\partial \beta_1} = -2\sum_{i=1}^{n} X_i(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$

Solving these yields: $\hat{\beta}_1 = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$

Appendix: Proof of Unbiasedness

Show: $E[\hat{\beta}_1] = \beta_1$ under assumptions 1-4.

Starting from $\hat{\beta}_1 = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}$

Substitute $Y_i = \beta_0 + \beta_1 X_i + u_i$:

$\hat{\beta}_1 = \beta_1 + \frac{\sum(X_i - \bar{X})u_i}{\sum(X_i - \bar{X})^2}$

Take expectations: $E[\hat{\beta}_1 | X] = \beta_1 + \frac{\sum(X_i - \bar{X})E[u_i|X]}{\sum(X_i - \bar{X})^2} = \beta_1 + 0 = \beta_1$

by the zero conditional mean assumption $E[u_i|X] = 0$.

Appendix: Variance of OLS Estimator

Under homoskedasticity:

$\text{Var}(\hat{\beta}_1|X) = \frac{\sigma^2}{\sum(X_i - \bar{X})^2} = \frac{\sigma^2}{SST_x}$

Key insights:

- Variance decreases with sample size (SST_x increases with n)
- Variance decreases with more variation in X (larger SST_x)
- Variance increases with larger error variance (σ^2)

Estimate σ^2 with: $\hat{\sigma}^2 = \frac{1}{n-k-1} \sum_{i=1}^{n} \hat{u}_i^2 = \frac{RSS}{n-k-1}$